

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
 BURP 4th Year 2nd Term Examination, 2017



URP 4275

(Development Planning)

Full Marks: 210

Time: 3.0 hrs

- N.B.: i) Answer any **three** questions from each section in separate script
 ii) Figures in the right margin indicate marks
 iii) Assume data if necessary

Section – A

Marks

1. (a) Define the concept of Development Planning. (05)
 (b) How will you differentiate "Central Planning" and "Indicative Planning"? Give appropriate examples. (15)
 (c) "Planning involves the formulation of a national program of action for achieving development objectives" - Justify the statement with example. (15)
2. (a) Write down the classification of Development Projects with approval authority. (10)
 (b) Write down the simplified series of steps in Investment Projects Processing/Approval Process. (10)
 (c) "Planning Commission is entrusted with the task of planning socio-economic development of the country" - Explain with proper example. (15)
3. (a) Write short note on Five Year Plan. Explain the major areas of improvement for higher growth during Sixth Five Year Plan (SFYP) with some examples. (20)
 (b) Make a short description on the background studies of Sixth Five Year Plan (SFYP) 2011-15. (15)
4. Write short notes on the following (answer any five): (35)
 (i) Gross National Happiness; (ii) Mention different approaches to the measurement of economic progress; (iii) Real GDP and Nominal GDP; (iv) ToR of ECNEC; (v) Input Output Model; (vi) Sectors/Divisions of Planning Commission; (vii) Functions of Physical Infrastructure Division

Section – B

5. (a) Define Development Planning. Explain Sen's "Capability" approach for development. (15)
 (b) Explain the rationale of Development Planning in the context of a third world country. (10)
 (c) Explain the concept of Prisoner's Dilemma as a reason of market failure. (10)
6. (a) Briefly describe the various methods of measuring Inequality. (10)
 (b) Calculate the Human Poverty Index (HPI) of Bangladesh from the information below. (15)
 Consider $\alpha = 3$.

Component	Value
P ₁	34%
P ₂	24%
P ₃₋₁	45%
P ₃₋₂	25%

- (c) Describe the range of policy options that can be taken to minimize Inequality and Poverty. (10)
7. (a) With the help of appropriate diagrams, clarify the Multiple Equilibria Approach. What are the problems of Multiple Equilibria? (15)
 (b) Describe the key assumptions and limitations of the Lewis Model. (10)
 (c) Briefly explain the Romer model and its criticisms. (10)
8. (a) Briefly describe the theories in the International Dependence Revolution. (15)
 (b) Annotate the following (any four): (20)
 (i) Endogenous Growth Theory; (ii) Agency Cost; (iii) Pareto Improvement; (iv) O-ring Theory with its implications; (v) Harrod-Domar model.

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 2nd Term Regular Examination, 2017



URP 4243

(Hazards and Disaster Management)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Draw your concept on Hazard, Disaster and Disaster Management Cycle. (08)
(b) Illustrate various phases of disaster management cycle with necessary example. (15)
(c) Illustrate the phrase - "Paradigm shift in disaster management" in the light of the experience of Bangladesh". (12)
2. (a) What do you mean by CBDRM? Mention the key characteristics of CBDRM. (08)
(b) Discuss in brief the steps of a community based disaster risk management program. (15)
(c) Why rapport building is necessary in a CBDRM program? What do you mean by participatory implementation of disaster risk management plan? (12)
3. (a) What do you know about the Stranding Order on Disasters (SOD) in Bangladesh? (08)
(b) Illustrate the Disaster Management Framework of Bangladesh. (12)
(c) What do you know about the structural and non-structural mitigation measures of disaster in Bangladesh? Discuss in brief. (15)
4. (a) Illustrate the objectives and priority areas of the Hyogo Framework for Actions (HFA 2005-2015). (08)
(b) Give a brief overview of the likely impacts of climate change in Bangladesh. (12)
(c) Highlight the salient features of the Bangladesh Climate Change Strategy and Action Plan 2009 as regards the impacts of climate change. (15)

Section – B

5. (a) "Disaster and National Image are inherently interrelated to each other" – Justify this statement in the light of immediate and long term impacts of disaster on a nation. (10)
(b) Explain hazard assessment approach with a focus to use of GIS. (10)
(c) "Risk is a combination of the interaction of hazard, exposure and vulnerability" – Write down your judgements to prove the statement. (10)
(d) Why is risk a multidisciplinary spatial problem? (05)
 6. (a) Write about the disaster management bodies in Bangladesh. (10)
(b) Classify various types of floods in Bangladesh. What are the deficiencies of flood management in Bangladesh? (10)
(c) Draw a stage-damage curve for flood hazard. What purpose is it used for? (07)
(d) Specify some measures that can be taken to strengthen Flood Forecasting and Warning Services (FFWS) in Bangladesh. (08)
 7. (a) Write down the potential applications of appropriate technologies in disaster management in the context of Bangladesh. (10)
(b) What components should a Cyclone Master Plan be considered? - Briefly explain with necessary example. (10)
(c) How is GIS used for risk assessment for different hazards? (10)
(d) How does psychology affect in coping with disaster? (05)
 8. (a) What do you understand by multiple hazards? Clarify with example. (05)
(b) What activities are covered presently under the Cyclone Preparedness Programme (CPP) taken by the Government of Bangladesh and the International Federation of Red Cross and Red Crescent Societies (IFRC)? (10)
(c) "Hazards create risk by exposing pre-existing vulnerabilities" – Explain the statement. (10)
(d) "Departments/organizations involved in development have no role in day-to-day disaster management activities" - Explain with necessary example. (10)
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Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 2nd Term Regular Examination, 2017
URP 4217
(Coastal and Special Zones Planning)



Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Compare the scientific definition of a coastal area with the policy oriented definition. (10)
(b) Illustrate the unique characteristics of coastal area. What are the significance/ importance of coastal area for a country having coast? (10)
(c) How could sustainability paradigm be used in coastal planning and management? Illustrate with necessary example. (15)
2. (a) Illustrate the difference between ICZM and conventional coastal zone management. (10)
(b) Discuss both the principles and strategies of ICZM. (15)
(c) Discuss the steps of ICZM plan formulation. (10)
3. (a) Discuss the effects of human pressures on coastal ecosystem. (08)
(b) Illustrate how public policies exacerbate the pressures on coastal environment. (10)
(c) Illustrate various tools and techniques that are used to manage growth in local level of coastal environment. (17)
4. (a) Why integrated conservation and development program is necessary for coastal conservation? Explain with necessary example. (10)
(b) Discuss various types of coast/coastal profile with necessary example. (15)
(c) Discuss the challenges of development of the coastal areas/zones of Bangladesh. (10)

Section – B

5. (a) Explain weathering and denudation. (10)
(b) "Coastal erosion and accretion is complex processes involving different factors" - show those factors in a flow diagram. (15)
(c) What are the differences between spring tide and neap tide? Discuss the importance of monitoring wave in the coastal areas. (10)
6. (a) Define coastal processes. (05)
(b) Explain how ocean and climatic factors contribute to coastal processes. (10)
(c) What are the different types of soil erosion? Mention different management strategies to control soil erosion? (10)
(d) Determine annual soil (silt loam) loss from an upland farm of 1000 acres in Khulna (Rainfall Erosivity Factor is 200). The slope length is 500 feet with 2 percent slope steepness (topographic factor is 0.326). The soil is covered with matured grass (0.004) and relative erodibilities of silt loam soils is 0.33. If slope steepness increases to 4 percent (topographic factor is 0.762), find out the changes in soil erosion. (10)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 2nd Term Regular Examination, 2017
URP 4217
(Coastal and Special Zones Planning)

Full Marks: 210

Time: 3.0 hrs

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Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 2nd Term Regular Examination, 2017
URP 4273
(Urban Governance)



Full Marks: 210

Time: 3 hrs

- N.B.** i) Figures in the right margin indicate full marks.
ii) Answer any **three** questions from each section in separate script.

Section – A

1. (a) Government is about “delegation of roles and responsibilities” and Governance is a “process” – Explain. Why is there a need for shift from government to governance? (10)
(b) How is informality and community involvement a key to governance system? (10)
(c) Civil society is an integral part of the neoliberal orthodoxy. Explain the problems of civil society in a developing country like Bangladesh. (10)
(d) What is the nexus between civil society and NGOs in Bangladesh? (05)
2. (a) Why do contemporary times demand for a Complex Adaptive System (CAS)? (10)
(b) What are the four major types of governance system? Explain with advantages and disadvantages. (15)
(c) Suppose a real-estate company wants to develop a new town close to Zero Point, Khulna. Identify the major actors involved in the whole process. If governance is a process, explain the negotiation process of the land and housing development. Do you think that such negotiation will bring the best-planned development? If ‘yes’, How? If ‘no’, why? (10)
3. (a) Define social development. Why is good governance in the forefront of global campaign for development and social development? (15)
(b) What are the major campaigns for good governance worldwide? What are the World Bank indicators for good governance? (10)
(c) What are the major problems of good governance in Bangladesh? (10)
4. (a) What is Cold War? With reference to chronological events, explain how fall of Berlin wall is symbolic to the end of cold war. (12)
(b) With reference to the backdrop of cold war, how do you understand the two parties - favoring and opposing - Bangladesh’s liberation war? (15)
(c) With reference to the global war and conflict, demonstrate that the tension of cold war seems to prevail in the present time. (08)

Section – B

5. (a) What are the objectives of measuring governance? (05)
(b) Explain the concept of Governance with relevance to Bangladesh. (10)
(c) Show Sherry R Arnstein’s “Ladder of Citizen participation”. (10)
(d) “Informal actors are more powerful than the formal actors” - Justify the statement (10)

6. (a) What are the issues of urban development? How are they related to governance? (10)
- (b) Briefly describe the modes of survey data collection. (10)
- (c) Which type of assessment is necessary for the following situations? Also mention the reasons behind your choice. (15)
- To find the level of corruption on a massive scale.
 - Applying PRA techniques.
 - Compare data between two or more countries.
 - To compute an input-output matrix for regional economics.
 - To apply Geo-Informatics on a particular topic.
7. (a) Show the differences between UGI and WGI. (10)
- (b) Calculate and explain the UGI for the following data: (10)
- Effectiveness sub-index = 0.54
 - Equity sub-index = 0.63
 - Participation sub-index = 0.47
 - Accountability sub-index = 0.91
- (c) How does a local government promote inclusive city development? (10)
- (d) A score of -0.45 WGI value and 15th percentile is achieved for control of corruption in Bangladesh after aggregating indicators. Provide your understanding of the results. (05)
8. (a) "Specific/local model" is most suitable for which of the governance measuring index? Describe that index. (15)
- (b) What are the objectives of Local Governance Self-Assessment (LGSA)? (05)
- (c) With the help of a spiral diagram show the following: (15)
- Issues of urban development
 - Break in spiral effect.
 - Urban development.

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 2nd Term Regular Examination, 2017
URP 4223
(Urban and Regional Economics)



Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
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iii) Assume data wherever necessary.

Section – A

1. (a) Why is regional economics important in urban and regional planning? (10)
(b) Suppose a URP, KUET graduate has two job options: (i) stay in Khulna and do freelancing; and (ii) go Dhaka and take up a job in a GIS firm. Considering per month standard living expense 10,000 BDT in Khulna and 15,000 BDT in Dhaka, explain his/her location choice based on the following condition. (20)

	Khulna	Dhaka
Income/month	25,000 BDT	30,000 BDT
One way transport (22 office days/month)	0	120
Social Cost (traffic jam-office days only, pollution, social network loss)	- traffic jam: 0 hrs - pollution: 5 BDT/day - social network loss: 10 BDT/day	- traffic jam: 2 hrs (25 BDT/hr) - pollution: 15 BDT/day - social network loss: 0 BDT/day
Opportunity Cost	100 BDT/day	0 BDT/day

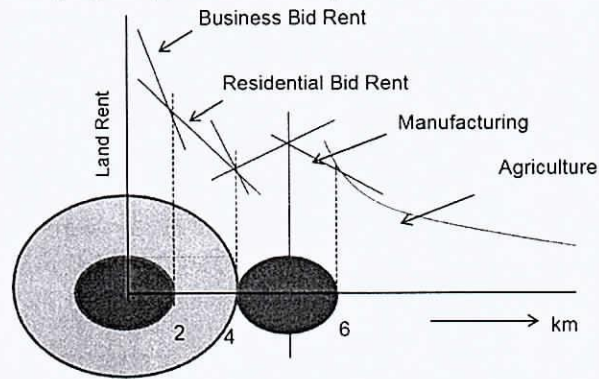
- (i) Explain the ideal location choice of the graduate.
(ii) (ii) In practice, what decision does the graduate make and why?
(iii) What are the impacts of such practical decision on the urban agglomeration of Dhaka or Khulna?
(iv) What urban and regional planning policies will you suggest for such a case?
- (c) Why and where do cities locate? (05)
2. (a) What is urban sprawl? What are the types of urban sprawl? What is the alternative to urban sprawl? Explain. (15)
(b) "21st century urbanization is urbanization of poverty and slummization" – Explain. (08)
(c) Explain the concept of Lorenz Curve and Gini-coefficient. Using the following data set, which (A, B) of the regions has the highest income inequality? Which one is more deserved by governments and urban and regional planners? (12)

A		B	
% Population	% Income	% Population	% Income
10	15	10	2
10	25	20	8
30	35	20	10
20	10	30	35
20	8	10	25
10	2	10	15

3. (a) What is bid-rent? What is the limitation of bid rent? (06)
(b) Using the following dataset, draw the bid rent of the city? (18)

	Production per land unit	Sell value per unit	Production cost per unit	Transportation cost per unit
A	50	10	8	0.2
B	45	9	7	0.1
C	40	8	6	0.05
D	45	5	3	0.025
E	30	3	1	0.02

- (c) Why do industries create a separate bid-rent beside the main city? Explain in reference to following figure, product life cycle and real-life examples. (11)



4. (a) Suppose a fast growing city has a high housing demand while it allows only 40,000 units per year and the equilibrium price per unit in the inner city area is 10 million BDT. Considering the demand, the city authority decides to expand housing supply by 30,000 additional units per year. Graphically explain the impact on the price change and estimate the monetary value of such permission. (15)
- (b) Following table shows four site options for an industry to locate in three different growth scenarios. Find out the equal probability pay-off, weighted pay-off and regret matrix. Which site will you recommend for the industry to locate and why? (10)

Payoff Matrix (in millions)			
Sites	Recession	Moderate Growth	Rapid Growth
Site 1	60	40	25
Site 2	30	45	60
Site 3	20	50	70
Site 4	20	67	30

- (c) Using Riley's law, calculate the market area of the following cities. What is the limitation of Riley's law explaining market area? (10)

Cities	Population	Distance from A
A	4,161	0
B	4,674	29
C	4,864	36

Section – B

5. (a) Narrate the relationship between Von Thünen's second model and land use pattern. (10)
- (b) Differentiate between cost minimizing and profit maximizing theories. Why Weber's Industrial Location theory is called a cost minimizing theory? (10)
- (c) Name a few of the basic and non-basic industries. (05)
- (d) "Shift-share analysis breaks down regional employment growth" – explain this statement with necessary examples. (10)
6. (a) The following table shows necessary information for industry i of a region for a nation. (15)

	Employment in Industry i	Total Employment	Industry Share of Area
Local Area	20	150	?
Local Area Share of Base Area	?	?	N/A
Base Area	500	10,000	?

- (i) Fill the blanks in the above table.
- (ii) Calculate basic employment in local area for industry i
- (iii) Calculate location quotient for industry i and decide whether it is self-sufficient, deficit or surplus?
- (b) Absolute advantage and comparative advantage theories support free trade in regional economics. Explain. (10)
- (c) Explain New Trade Theory with its implications (10)

7. (a) Assume you are working as a planner. You have one market place (A) with three materials sources (B, C and D) in your planning area. You can only chose the market and two material sources at a time. According to the following information what will be the possible arrangement of your material sources and market from your industry. (15)

	Point Specific Weights for import/export (ton)	Transport tariff/unit of weight/unit of distance (BDT)	Distance from Z (km)
A	300	50	370
B	200	40	250
C	250	60	400
D	175	35	350



- (b) Which growth factors influence the regional economic growth in developing country like Bangladesh? (10)
- (c) Illustrate the circular flow of income model. (10)

- 8 (a) The following table shows different industry's employment of a region for a nation along with national growth rate (12.8%), industry sector's national growth rate and regional industry's growth rate. (20)

Industry	Regional Employment (1993)	Industry's National Growth Rate (%)	Regional Industry's Growth Rate (%)
Farm	7,951	-0.1	0.3
Manufacturing	58,516	4.6	4.6
Retail	44,752	13.8	12.5
Finance and Real Estate	16,193	16.5	14.5
Service	62,518	19.3	20.7
All other	63,533	9.7	10.0

- (i) Calculate national growth share (NS), regional growth share (RS) and industry mix (MI) separately for each industry.
- (ii) Find out the fast growing and declining industries.
- (b) "A country may produce goods more efficiently because of a natural advantage or because of an acquired advantage" - Discuss this statement with necessary examples. (10)
- (c) Draw simple input-output flow table for one area. (05)
